

(TR-104) PYTHON WITH AI/ML

Training Day 47 Report

Date: 17 March 2026

Topic Covered: RAG Pipeline Configuration Optimization & Revision of Core Concepts

The forty-seventh day of training focused on optimizing configuration handling in the RAG-based AI Assistant project and revising important concepts related to LangChain, document processing, embeddings, vector databases, and retrieval pipelines.

- Improved project configuration by increasing API rate limit settings for better request handling performance.
- Moved hardcoded values such as cache TTL and rate limiting configurations into environment variables and centralized configuration files.
- Strengthened project maintainability by reducing hardcoded parameters inside the application.
- Revised the purpose and working of LangChain components used in RAG systems.
- Reviewed document loaders and understood how different file formats are processed and converted into text data.
- Revised Text Splitters and analyzed how large documents are divided into smaller semantic chunks.
- Reviewed the complete RAG pipeline workflow from document ingestion to final answer generation.
- Studied how embeddings are generated from text and converted into vector representations.
- Revised the role of vector databases such as ChromaDB in semantic retrieval systems.
- Reviewed retrieval logic and similarity-based document searching techniques.
- Revised how retrieved context is passed into Large Language Models for response generation.

- Strengthened understanding of query handling and retrieval flow for accurate context-aware answers.
- Improved conceptual understanding of data ingestion pipelines and retrieval pipelines in production-level RAG systems.
- Strengthened overall understanding of RAG architecture, retrieval optimization, and context-grounded AI response generation.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/rag_pipeline
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/loading-and-processing-documents-with-langchain-a25a0e812cf7>